



Voice from IIT Kharagpur

“Everybody knows that knowledge is power and technology is power multiplier”



Professor Damodar Acharya, director, IIT Kharagpur, has about 32 years of diversified experience—27 years of teaching and research experience at IIT, Kharagpur, and five years of experience in academic administration as vice-chancellor of Biju Patnaik University of Technology, Rourkela, and chairman of All India Council of Technical Education (AICTE), New Delhi.

Recently, Prof. Acharya spoke to EFY's research analyst Sudeshna Das about their use of technology to strengthen the teaching-learning ecosystem

Does technology empower teaching?

The answer is an unequivocal 'yes.' Everybody knows that knowledge is power and technology is power multiplier.

This fact is also very well recognised by the government of India. The MHRD has launched a major project called 'National Mission Project on Education through Information & Communication Technology' with a budget of Rs 100 billion during the Eleventh Plan period.

Initiatives taken by IIT Kharagpur

IIT Kharagpur is the only IIT in the country with a full-fledged Centre for Educational Technology (CET) offering both technology courses related to education (M.Tech in media & sound engineering) and courses in hearing science ('instructional system design' ICT in education).

This centre was established way back in 1987 but since 1995 it has been recognised as the leading centre of its kind in

the country. CET, IIT Kharagpur, has been conducting major R&D activities in the area of technology-enhanced hearing for nearly 15 years now. It developed and experimented with the use of virtual classroom as far back as 2001-02 within our campus, where anyone connected to our LAN could interactively (through one-way video and two-way audio and text) participate in any of our video classrooms.

We were the first and for many years the only institute in India, and perhaps one of the handful of higher educational institutions in the world, to introduce video-on-demand (VOD) facility in the campus. The VOD service was started by CET in 2001 immediately after our LAN infrastructure was upgraded. This allowed anyone connected to our LAN to access video-recorded courses from anywhere, anytime with VCR-like functionalities. In 2001, when we had only 23 courses (each of 40-hour duration), the average daily hit rate was 3800-4000. Today, we have nearly 200

40-hour courses.

We had introduced, and I must say very successfully, video as a medium of regular instruction way back in 1996, when very few people were even aware of the possibilities. Unlike the usual enrichment-type TV programmes in use at that time produced by UGC/EMRC/IGNOU/NCERT, we decided to produce curriculum-based full courses (typically 40x1-hour long).

IIT Kharagpur's unique EInet 3L (ELectronically Networked Life Long Learning) experiment in using technology-enhanced learning for increasing the reach of IIT Kharagpur faculty to the society at large is still considered a landmark experiment in technical education. This 'tutored video instruction with small group learning' concept trained over 12,000 people, through 109 study centres spread all over India, in nine ICT subjects. The experiment lasted five years—from 1996 to 2001. This was a major innovation of the use of technology in education.